cylinders that are kept separately on the back plates.

When the brake pedal is pressed, the brake fluid inside the master cylinder is pressurized and sent to the wheel cylinders through four pipe lines and pushes the pistons inside the cylinder in outward direction. At that time the brake shoes which are connected to the pistons get expanded and force the rotating brake drum to stop which stops the rotation of the wheel.

When the brake pedal is released the pressure of the brake fluid in the master cylinder is reduced. This causes the piston in the master cylinder to come back to its original position due to the spring's action. Thus, pressure in the brake lines is reduced completely. Due to the drop in pressure of the brake fluid, the piston inside the wheel cylinder is move inward. Due to this action, the brake shoes get contracted by the return spring and moved away from the brake drum. Thus, the wheels are allowed to rotate freely. By this way, returning action of piston in the wheel cylinder makes the brake fluid to return back to the master cylinder.

Merits and demerits of hydraulic brake system

The following are the merits and demerits of the hydraulic braking system used in automobiles.

Merits

1. Unlike the mechanical brakes it is simple in construction with no mechanical lever and linkage.
2. The braking force in this system is uniformly distributed to all the wheels.
3. Parts of this system are self-lubricated.
4. It occupies less space.
5. More braking efficiency could be achieved.
6. The braking force is equally shared for all the four wheels.
7. Even for small force of the driver on the brake pedal, the master cylinder multiplies the pressure and gives increased mechanical advantage.
8. Due to the possibility in varying the diameter of the brake pipe lines from the wheel cylinders, variable braking efficiency could be achieved.

Demerits

1. Even a small leakage in the brake lines makes the entire system failure.
2. The air entering into the system reduces the braking efficiency.
3. There is a chance of oil leakage from wheel cylinder. Due to this the frictional force and braking force get reduced in between brake drum and brake shoe.

Quality of the brake fluid

The brake fluid is an important part which acts as a brake force transmitter in the hydraulic braking system. The compression of the brake fluid increases the pressure in the system. However the volumetric capacity of the fluid does not change. The pressure applied at one point on the fluid is equally distributed in all directions, based on the scientific law Pascal by that way the hydraulic braking system works. The brake fluid is generally made up of glycol with different alcoholic additives. In addition, the brake fluid can also be prepared by good quality silicon based fluids.

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